
Think or Talk?

Post-Training Expert Routing for Real-Time Duplex Speech Models

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 A real-time duplex speech model should answer easy questions immediately, but
2 hard questions often benefit from waiting for stronger reasoning. The challenge
3 is to choose between these paths at the moment the model begins to speak. Re-
4 cent systems learn related mode decisions through special-token generation inside
5 the dialogue model, coupling routing to turn-taking and temporally annotated full-
6 duplex training. We study a post-training alternative. We freeze MiniCPM-o 4.5
7 and fit a logistic failure readout on hidden states already available at its native
8 listen-to-speak transition. The resulting gate requires no additional model genera-
9 tion and exposes thresholds that can be retuned without changing the duplex policy.
10 The signal learned from text transfers to unseen human speech, while native-state
11 aggregation improves its reliability. On the internal benchmark, selective hand-
12 off improves answer accuracy by 22.1 points. Compared with always using the
13 expert, it reduces mean server time to first answer audio by 35.3 percent. It also
14 outperforms matched random routing across external speech question-answering
15 pools and transfers to a second duplex model family. These results show that an
16 existing duplex model can be equipped to answer easy turns immediately and wait
17 for stronger reasoning on harder ones, without another round of duplex training.

18 1 Introduction

19 Real-time conversation rewards two behaviors that pull in opposite directions. For a greeting or an
20 easy factual question, a speech agent should answer as soon as it understands the turn. For a difficult
21 knowledge or reasoning question, speaking immediately can lock the interaction into a fluent but
22 incorrect answer; taking time to reason or ask a stronger model is preferable. The choice is therefore
23 whether to answer now or wait for stronger reasoning.

24 Native duplex models already learn a fine-grained listen/speak policy [Défossez et al., 2024, Cui
25 et al., 2026]. Recent systems extend this idea with tokenized control. BayLing-Duplex uses
26 four dialogue-state tokens for turn-taking and interruption [Fang et al., 2026]; DuplexOmni uses
27 a [THINK] token to request an asynchronous reasoning layer [Huang et al., 2026]; and Listen-Write-
28 Speak encodes concurrent listening, writing, and speaking with a token schema [Zhang et al., 2026].
29 These systems show that special-token control is expressive. They also make the routing rule part
30 of autoregressive behavior: the model must learn rare control events together with speech content,
31 timing, and interruption. BayLing-Duplex, for example, reweights rare state tokens, adds timing-
32 focused preference optimization, and retains an auxiliary language-modeling loss. DuplexOmni
33 and Listen-Write-Speak construct large temporally annotated training corpora. Updating the rule
34 consequently means updating or retraining the interaction model.

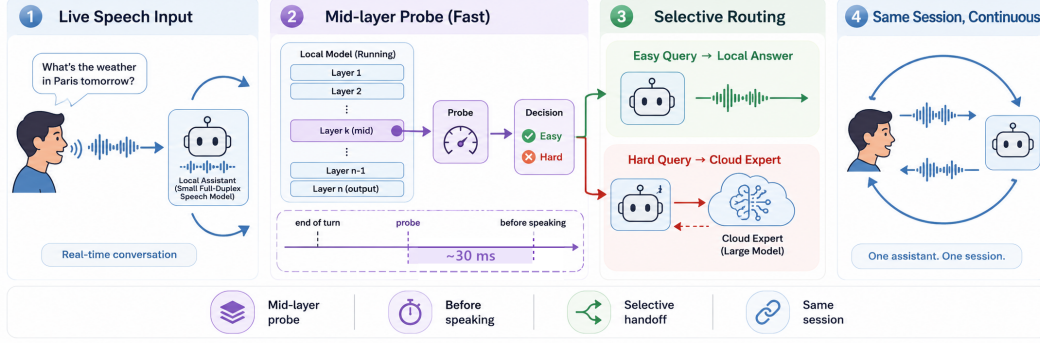


Figure 1: **Post-training routing at the native speak boundary.** (1) Speech streams in 1 s chunks to a frozen full-duplex local model. (2) When the model commits to speak, an L22 linear readout scores local failure risk in tens of milliseconds; the triggering call may already contain the first text unit. (3) Low-risk turns continue immediately, while high-risk turns are sent to a stronger expert and the talker holds the floor. (4) The expert answer is voiced by the local talker. On all 240 internal queries, the aggressive policy raises accuracy from 40.8% to 62.9% at 51.7% escalation.

Turn-taking supervision also does not directly answer the competence question. A model may be ready to take the floor while lacking the knowledge needed for a correct response. Conversely, routinely invoking a strong expert wastes cost and makes easy turns wait. What is needed is a failure-risk estimate that is available at the native speak boundary, does not disturb the learned duplex policy, and can be recalibrated as the desired accuracy–latency operating point changes.

We investigate whether this decision can be recovered from a frozen duplex model’s own representations. Hidden-state probes can predict correctness in text models [Azaria and Mitchell, 2023, Kossen et al., 2024, Lugoloobi et al., 2026], but a real-time speech setting introduces three shifts. The input is audio rather than text; duplex fine-tuning reshapes representations for interaction; and the read must be available at the listen-to-speak transition, before a complete answer can be inspected. A successful gate must survive all three and produce a useful serving policy, not merely an offline AUC measured after the answer has already been generated.

Our system freezes MiniCPM-o 4.5 [Cui et al., 2026] and trains only a regularized logistic readout on cached answer-onset states. This separates *interaction control*—when the model takes the floor—from *competence control*—whether its answer should be trusted. The readout adds no language-model forward pass, leaves local responses unchanged, and turns the expert-call budget into a threshold that can be adjusted after training. On the expert path, the question is sent to a stronger model and the returned answer is spoken by the local talker.

Our approach treats competence as a post-training serving decision. We implement think-or-talk routing at the native speak boundary with a linear gate, without adding control tokens, updating the backbone, or repeating duplex training. We trace the readable signal through duplex and modality shifts: text-trained probes transfer to unseen human speech, while the matched native ablation identifies answer-onset aggregation as important for robust readout. We evaluate the gate as a complete serving policy rather than only an offline classifier. On the 240-query internal set, selective handoff improves accuracy by 22.1 points while reducing mean server waiting time by 35.3% relative to always escalating. Across four external speech-QA pools, the aggressive policy improves macro accuracy by 21.4 points over local answering and by 5.9 points over a matched random router at the same expert-call rate in every external evaluation pool.

The result is a practical division of labor: the duplex model preserves the fast interaction for which it was trained, while a separately calibrated gate decides when additional reasoning is worth the delay.

2 Related Work

Learned control in duplex speech models. Native spoken-dialogue models coordinate concurrent listening and speaking [Défossez et al., 2024, Cui et al., 2026], using learned states, synchronization mechanisms, or lightweight adapters [Wang et al., 2024a, Zhang et al., 2024, Veluri et al.,

2024, Wang et al., 2024b]. BayLing-Duplex interleaves user audio, assistant text, and assistant audio; four special tokens reduce silence, reply onset, text completion, and speech completion to next-token prediction [Fang et al., 2026]. MiniCPM-o 4.5 similarly predicts a binary Listen/Speak token before content. These signals control the floor, not whether the forthcoming answer is correct.

Reasoning while interacting. Speech models have been designed to reason before speaking, interleave reasoning with speech, or reason while listening [Wu et al., 2026]. Listen-Write-Speak uses a token schema for an always-on visible writing channel alongside listening and speaking [Zhang et al., 2026]. DuplexOmni is the closest control-based handoff: its interaction model emits [THINK] to invoke a pluggable background reasoner and learns delayed-result integration from Writer-Director data [Huang et al., 2026]. MoshiRAG instead triggers retrieval from the delay between speech onset and the first content word, then gives documents back to the local model [Chien et al., 2026]. Our gate reads failure risk from the frozen model at answer onset, before a completed local answer, and delegates the answer to an expert. It can therefore be added to an existing duplex checkpoint and retuned without relearning its dialogue policy.

Competence readout and selective routing. Correctness can be estimated from verbal self-evaluation [Kadavath et al., 2022, Lin et al., 2022], hidden representations [Azaria and Mitchell, 2023, Burns et al., 2023, Marks and Tegmark, 2024], or uncertainty across sampled answers [Kuhn et al., 2023, Farquhar et al., 2024]. Intermediate layers can retain information obscured at the output [Orgad et al., 2025, Mahaut et al., 2024], and Semantic Entropy Probes amortize sample-based uncertainty into a hidden-state read [Kossen et al., 2024]. Text routers allocate queries among models of different cost [Chen et al., 2023, Ding et al., 2024, Ong et al., 2024], including from prefill activations [Varshney et al., 2026, Lugoloobi et al., 2026]. A linear probe is therefore not our novelty. Our contribution is to make such a readout work at a native audio listen-to-speak boundary, study its survival across speech and duplex transformations, and measure the resulting accuracy-latency policy in a live speech pipeline with end-to-end expert handoff.

3 Post-Training Think-or-Talk Routing

3.1 The local-expert loop

Selective model routing balances response quality against the cost of a stronger model [Ong et al., 2024]. We place that decision at a local full-duplex speech model’s transition from listening to speaking (Figure 1). The local model retains its native decision about when to respond; the gate decides whether the response should begin immediately or wait for expert help. On the expert path, an audio uplink transcribes the question, the expert returns a text answer, and the local talker voices the prepared relay. We use predicted local-answer failure as the routing score and vary a threshold to expose the accuracy-latency frontier across different expert-call budgets.

3.2 A probe for the handoff decision

Linear *probes* test what information a classifier can read from frozen representations [Alain and Bengio, 2016]. Hidden-state classifiers have estimated statement truthfulness [Azaria and Mitchell, 2023] and pre-generation task success [Lugoloobi et al., 2026]. We use this approach to ask whether a speech model’s existing states contain a directly readable failure signal. Reusing these states avoids a separate self-evaluation generation and leaves the speech model unchanged.

Offline, we collect local answers to calibration questions and use a judge to assign correctness labels. Each label is paired with hidden-state features captured at the chosen read point. We fit a logistic probe to predict failure from these features while keeping the speech model frozen. For features $\phi(x)$ from input context x , the fitted weights w, b give the risk score

$$s(x) = \sigma(w^\top \phi(x) + b). \quad (1)$$

At inference, the probe reads the current features without a reference answer or an online judge. Scores above a language-specific threshold τ_ℓ favor escalation; thresholds select nominal expert-call budgets from calibration scores and stay fixed during evaluation (§4.1). Realized call rates can change when the query distribution shifts. An auxiliary dialogue-act head restricts escalation to information requests, filtering turns such as acknowledgments and stop commands (Appendix D).

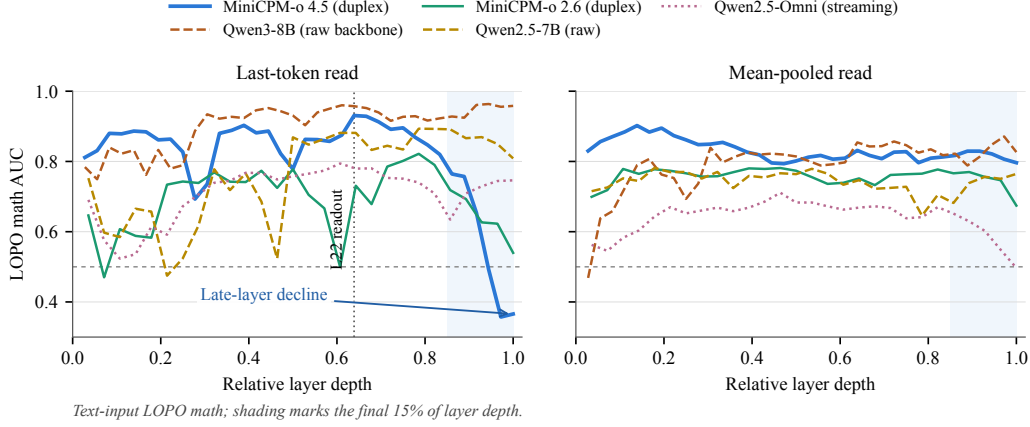


Figure 2: **Text-input transfer depends on layer and position.** The arrow and shading highlight MiniCPM-o 4.5’s late-layer decline in text-input LOPO math AUC. MiniCPM-o predicts a binary Listen/Speak control token before content [Cui et al., 2026, §3.3], but direct controls do not identify turn control as the cause of the decline. L22 is the native readout; §4.4 and Appendix B.2 report the corresponding audio-input results and mechanism checks.

118 **Why separate routing from generation?** Only w and b are fitted; the regularized logistic ob-
 119 jective is convex and uses cached features and answer labels. It therefore cannot alter the local
 120 model’s answer distribution, listen/speak behavior, or speech quality. The operating threshold is a
 121 serving parameter rather than a generated token, so it can be recalibrated for a new expert budget
 122 without another duplex fine-tune. This is the precise training-stability advantage of our design: it
 123 removes competence arbitration from the joint autoregressive objective rather than claiming that
 124 special-token control cannot be trained successfully.

125 3.3 Where should the probe read?

126 Adapting the probe to duplex speech requires choosing its layer, position, and read time. The final
 127 layer is optimized for the output head, whereas an earlier layer can retain more transferable task
 128 information. Figure 2 motivates L22 through text-input leave-one-pool-out (LOPO) evaluation: on
 129 held-out math, its last-token probe reaches 0.931 AUC, versus 0.366 at the final layer. The contrast is
 130 regime-specific rather than a universal middle-layer advantage: it disappears when both calibration
 131 and inference use audio input in the matched native setting (§4.4).

132 The native feature vector concatenates the latest L22 tail state, the mean of the tail states, and the
 133 mean input state, allowing the probe to combine the current boundary state with pooled context.
 134 The matched native ablation in §4.4 shows that this aggregation matters: a single L22 state does not
 135 outperform a single final-layer state. Both depth and aggregation therefore define the native readout.

136 We read these features when the generation call first returns a listen-to-speak transition. This makes
 137 the gate available at the local model’s response boundary without waiting for a completed answer.
 138 The call can already contain the first generated text unit, so we call the decision point *answer onset*.
 139 Further answer chunks may expose additional failure information, at the cost of a later handoff
 140 (Appendix D.3). We fit the native probe and thresholds on features and answer labels from this
 141 serving regime to match its audio input and generation behavior. This calibration is separate from
 142 the representation-transfer study (§4.5), which alone does not provide a zero-shot native gate.

143 4 Experiments

144 4.1 Experimental setup

145 We evaluate frozen MiniCPM-o 4.5 (9B), with gpt-5.5 as the escalation expert, on the full 240-
 146 query internal test set and four external English speech-QA pools: Speech TriviaQA, Speech Web
 147 Questions, and Speech Llama Questions from OpenAudioBench [Li et al., 2025], and SD-QA from

Table 1: **Answer-content accuracy (%)**. Both models use 240 Internal IDs and four external QA pools; avg is the external mean and Δ its gain over local. MiniCPM uses native sessions; VoiceChat-11B uses cached replay across three routing budgets. Random rows match each aggressive call rate.

	Internal	TriviaQA	WebQ	Llama Q.	SD-QA	avg	Δ
<i>MiniCPM-o 4.5 native sessions</i>							
always-local	40.8	62.8	52.8	82.4	48.0	61.5	—
+ gate, conservative	46.7	60.4	59.2	78.8	54.0	63.1	+1.6
+ gate, balanced	54.2	70.0	62.4	82.4	65.0	70.0	+8.5
+ gate, aggressive	62.9	88.4	75.2	85.6	82.5	82.9	+21.4
matched random, aggressive	56.1	81.5	68.9	84.1	73.7	77.1	+15.6
always-escalate	70.4	96.0	79.6	91.6	87.0	88.6	+27.1
<i>VoiceChat-11B cached native-frame replay</i>							
always-local	19.6	35.8	37.6	70.8	31.0	43.8	—
+ gate, conservative	31.7	48.4	48.8	80.7	42.0	55.0	+11.2
+ gate, balanced	44.6	61.0	59.2	85.8	54.0	65.0	+21.2
+ gate, aggressive	60.4	76.8	69.6	89.7	66.5	75.7	+31.9
matched random, aggressive	53.5	65.7	60.2	82.0	60.0	67.0	+23.2
always-escalate	91.7	95.5	82.8	93.1	89.0	90.1	+46.3

VoiceBench [Faisal et al., 2021, Chen et al., 2024]. For the answer-content benchmark, the native probe is fitted on 5,221 calibration rows after excluding benchmark overlaps. Its results were rerun with this refitted probe; per-language thresholds use the remaining core rows’ out-of-fold scores and stay fixed across pools. Calibration details and model controls are in Appendix A.

Each query has one sampled native session per arm: local-only, conservative, balanced, aggressive, and always-escalate. We report *answer-content accuracy*, scored from generated local text or intended relay text. The random reference is the analytical mixture $A_{\text{mix}}(r) = (1-r)A_{\text{local}} + rA_{\text{always}}$ at the gate’s realized call rate r . Both model families use all 240 Internal query IDs: chat (60), factual QA (40), knowledge (60), math (60), and SimpleQA traps (20). Every query has unit weight in accuracy and call-rate calculations. Table 7 gives the per-tier results and uncertainty intervals on this full test set. External averages weight the four QA pools equally.

Metric sources and timing endpoint. All tables and trade-off figures use the retained judged benchmark for accuracy and escalation rate, and the complete ttfa-v3 run for latency on the same query IDs. This timing run contains 5,950 recorded sessions across five pools and five policies; TTFA comes directly from session timestamps. We compare the resulting per-policy summaries. Appendix A.2 gives the input and recording details for each metric. Server TTFA measures the wait from scheduled input end until the first answer PCM block is ready; audio ready earlier incurs zero wait. Candidate and stall audio are excluded; client playback is unmeasured. Appendix A.1 provides the model, probe-fitting, and answer-judging configurations used in these experiments.

4.2 Internal accuracy and timing cost

The upper MiniCPM block of Table 1 reports unweighted answer-content accuracy on the full 240-query internal test set: 40.8% locally, then 46.7%, 54.2%, and 62.9% for conservative, balanced, and aggressive. Their realized escalation rates are 8.3%, 25.0%, and 51.7% (Appendix D.1). Aggressive exceeds the matched random reference of 56.1% by 6.8 percentage points, computed before rounding. Per-tier results and accuracy intervals are in Table 7. Aggressive recovers 74.6% of the measured local-to-always accuracy gap at a call rate of 51.7%. The gain is concentrated on factual QA and questions with misleading premises, where the gate can identify that immediate local speech is risky; all queries and per-category results are retained in Appendix A.3.

The complete ttfa-v3 run measures server waiting time to first answer audio after input end on the same 240 query IDs (Figure 3). Internal P50 is 0.70 s locally, 1.61 s at balanced, and 2.87 s at aggressive. Aggressive has mean 6.10 s and P95 18.14 s; always has mean 9.43 s and P50 6.67 s. Local mean is 1.06 s, including 17/240 attempts (7.1%) that respond before input end and incur zero wait. Early responses do not establish better interaction or correctness. Table 8 counts failures and reports timing among completed sessions. Under the reporting convention in §4.1, aggressive has 62.9% accuracy and 6.10 s mean TTFA, compared with 70.4% and 9.43 s for always-escalate. Thus

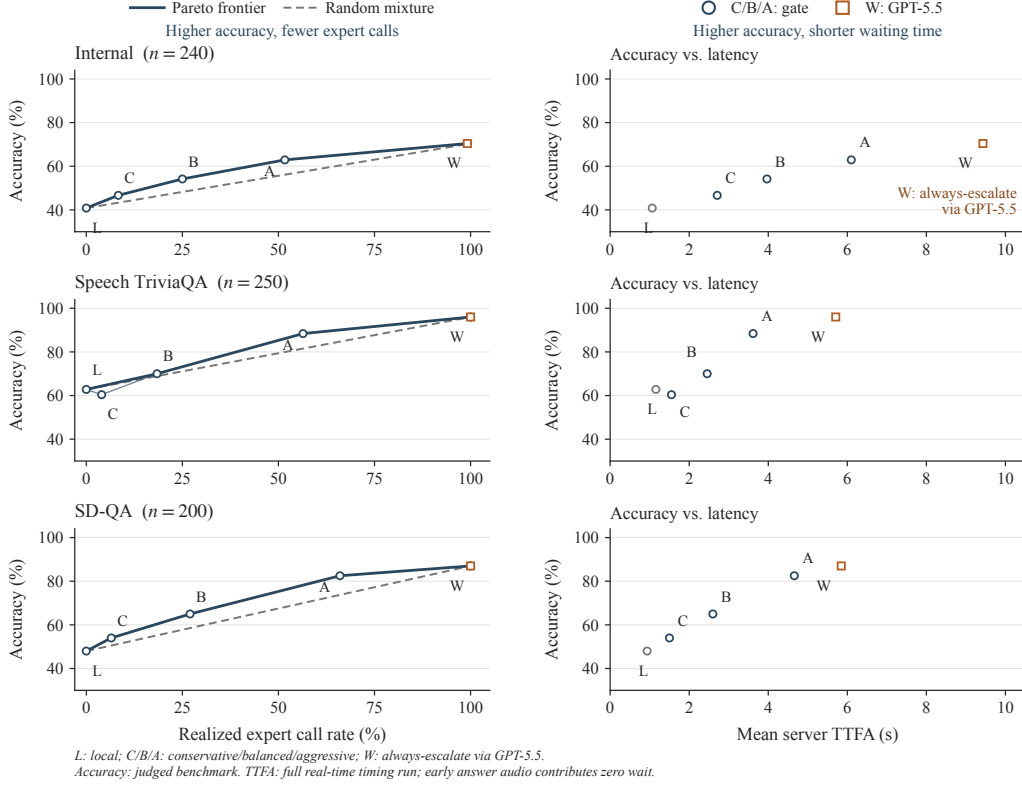


Figure 3: **Accuracy–cost trade-offs across routing policies.** Left: accuracy versus expert call rate, with thin policy curves, thick Pareto frontiers, and dashed random-mixture lines. Right: accuracy versus mean server TTFA, shown as unconnected points. Always-escalate via GPT-5.5 (W) has the highest accuracy and latency on these pools; gated policies trade some accuracy for shorter waits. TriviaQA conservative is dominated by local. Accuracy retains the judged benchmark values; TTFA uses the full timing run (§4.1). Internal retains all 240 IDs with unit weights. L/C/B/A/W denote local, conservative, balanced, aggressive, and always-expert policies.

183 the post-training gate retains 89.3% of always-expert accuracy while reducing mean waiting time by
 184 35.3%, without retraining or modifying the duplex generation policy.

185 4.3 External transfer and its limits

186 The native gate’s weights, feature definition, layer, and per-language thresholds stay fixed across the
 187 four external English QA pools. The *avg* column excludes the internal set and Δ is its gain over
 188 always-local. Aggressive macro accuracy rises from **61.5% to 82.9%**, compared with 77.1% for
 189 the matched random policy: +21.4 points over local answering and +5.9 over random. Realized
 190 call rates adapt to the pool, from 18.8% on Llama Questions to 66.0% on SD-QA. TriviaQA and
 191 SD-QA show the largest gains over local answering (+25.6 and +34.5 points) and over their random
 192 references (+6.9 and +8.8 points). The lower-budget tiers make smaller changes, as expected when
 193 the expert is reserved for only the highest-risk tail.

194 Failure categories and later reads help explain the boundary. In the native failure-type audit, specific-
 195 but-wrong answers comprise 70% of failures and are ranked at AUC 0.81, while execution slips in
 196 multi-step reasoning are near chance (0.47). In a separate read-time sweep, external mean AUC
 197 increases from 0.755 at commitment to 0.791 after two answer chunks. Later chunks expose useful
 198 failure information without identifying the mechanism underlying this change (Appendix D.3).

199 The lower block of Table 1 provides a portability check on NemotronLabs-VoiceChat-11B. Its
 200 three-layer answer-onset readout reaches 0.818 out-of-fold and 0.816 mean external AUC. At a

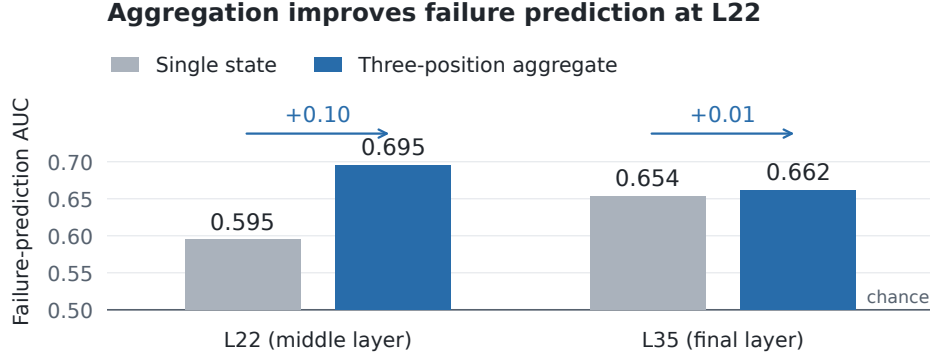


Figure 4: **Aggregation improves failure prediction at L22.** Mean AUC across five external pools under leave-one-pool-out evaluation on one native audio batch. Gray bars read the latest tail state; blue bars concatenate that state with the tail mean and input mean (§3.3). Arrows indicate the aggregation gain at each layer; the horizontal line marks chance AUC of 0.5.

201 50% cached replay budget, external macro accuracy rises from 43.8% to 75.7%, versus 67.0% for
 202 matched random routing. All tiers appear in the table; Appendix C gives replay details.

203 4.4 Readout ablations

204 **Text-input transfer.** We evaluate the readout without using expert outcomes. The sweep in Fig-
 205 ure 2 compares read positions and raw-backbone controls; mean pooling reduces late-layer degrada-
 206 tion. A raw Qwen backbone, subsampled to match per-pool failure rates, remains near 0.96, with
 207 similar ordering in the Qwen2.5 control family (Table 3). With audio calibration and audio input,
 208 the final-layer probe reaches 0.936 on LOPO math. These controls link degradation to the tested
 209 speech checkpoints and input regimes without identifying its cause (Appendix B.2).

210 **Native layer and aggregation.** We test whether predictive gains at native answer onset come from
 211 layer choice or feature aggregation. A 2×2 comparison crosses L22 versus the final layer with a
 212 single tail state versus the three-part aggregate defined in §3.3. All four readouts use one audio batch
 213 with identical training pools, labels, splits, and head recipe (Figure 4). At L22, aggregation raises
 214 mean failure-prediction AUC from 0.595 to 0.695; at the final layer, the change is smaller (0.654
 215 to 0.662). The single-state L22 read has no advantage over the final layer. These results identify
 216 an aggregation benefit at L22, without establishing an advantage from middle-layer selection alone.
 217 Named per-pool results and cached routing gains are reported in Appendix B.3.

218 4.5 Text-trained probes transfer to unseen human speech

219 We next test whether a probe fitted to text-input hidden states can predict failure under speech input.
 220 This tests a change of input modality; Figure 2 tests held-out task-pool transfer within text input.

221 Figure 5 (left) holds question content fixed: the same 600 queries are presented as text and as TTS
 222 audio. The text-trained MiniCPM-o 4.5 probe reaches **0.855** AUC on audio states at L22. Transfer
 223 also appears in MiniCPM-o 2.6 and Qwen2.5-Omni, with plateau AUCs of 0.74–0.80 and 0.80–
 224 0.83, respectively; the frozen-backbone Freeze-Omni control shows much weaker transfer under the
 225 tested recipe. This panel tests modality transfer with shared question content.

226 The right panel tests generalization to *new questions and human speech*: probes fitted on the calibra-
 227 tion pool are evaluated on 200 query-disjoint SD-QA recordings. The text-trained MiniCPM-o 4.5
 228 probe reaches **0.760** AUC at L22, with a per-layer peak of 0.800 at L25. Audio-trained probes
 229 and evaluation on SD-QA text provide references for the modality change. Thus, transfer extends
 230 beyond matched TTS questions to held-out human speech. The two panels use different evaluation
 231 sets, so their AUC difference does not isolate a human-speech penalty (Appendix B.1). These results
 232 concern representation transfer; the native gate requires a separately fitted readout (§3.3).

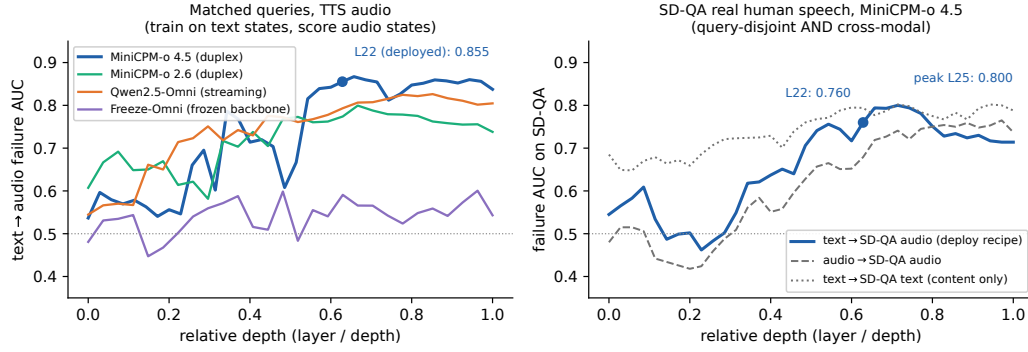


Figure 5: **Text-trained failure probes transfer to unseen human speech.** Left: text-trained probes are evaluated on TTS audio for the same 600 questions across four models; MiniCPM-o 4.5 reaches 0.855 AUC at L22. Right: MiniCPM-o 4.5 probes fitted on the calibration pool are evaluated on 200 query-disjoint SD-QA human recordings. The text-trained probe (blue) reaches 0.760 at L22 and peaks at 0.800 at L25; gray curves show audio-trained and text-target references. Both panels show relative depth, normalized by the total number of layers in each model.

5 Discussion

The central design choice is separation, not a new way to train a duplex model. A learned Listen/Speak or [THINK] token is useful when the desired policy and training distribution are fixed. A post-training readout is useful when the interaction checkpoint should remain unchanged and the cost of waiting or calling an expert may vary after deployment. Because the readout uses answer-onset states already produced by the model, the marginal local computation is small; because the threshold is external, the operating point can change without modifying generation.

The results also distinguish floor readiness from answer readiness. The native model’s decision to speak supplies a well-timed intervention point, but a separate competence signal is needed to decide whether taking that turn is wise. Mid-layer transfer and native aggregation are two complementary ways of preserving that signal under distribution and modality shift.

Our evidence covers two duplex model families and primarily single-turn speech QA. The gate uses in-regime labeled calibration, and TTFA ends at server PCM readiness. Multi-turn continuity and human-rated interaction quality remain important next evaluations.

6 Conclusion

Real-time speech agents should neither delay every easy turn nor answer hard turns without sufficient reasoning. We show that this choice need not be learned as another generated control token. Failure signals remain readable in frozen duplex-model states, transfer to unseen human speech, and enable expert handoff at the native speak boundary of a live conversational pipeline.

The architecture keeps fast interaction intact while moving competence arbitration into a lightweight, tunable serving layer. This yields an adjustable accuracy–latency frontier rather than a fixed think-or-talk behavior. Full-duplex control should govern both *when the agent has the floor* and *when it should trust itself enough to use it during an ongoing real-time interaction*.

References

- Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes. *arXiv preprint arXiv:1610.01644*, 2016.
- Amos Azaria and Tom Mitchell. The internal state of an LLM knows when it’s lying. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023. arXiv:2304.13734.
- Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. Discovering latent knowledge in language models without supervision. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2212.03827.
- Lingjiao Chen, Matei Zaharia, and James Zou. FrugalGPT: How to use large language models while reducing cost and improving performance. *arXiv preprint arXiv:2305.05176*, 2023.
- Yiming Chen, Xianghu Yue, Chen Zhang, Xiaoxue Gao, Robby T. Tan, and Haizhou Li. VoiceBench: Benchmarking LLM-based voice assistants. *arXiv preprint arXiv:2410.17196*, 2024.
- Chung-Ming Chien, Manu Orsini, Eugene Kharitonov, Neil Zeghidour, Karen Livescu, and Alexandre Défossez. MoshiRAG: Asynchronous knowledge retrieval for full-duplex speech language models. In *International Conference on Machine Learning (ICML)*, 2026. arXiv:2604.12928.
- Junbo Cui, Bokai Xu, Chongyi Wang, Tianyu Yu, Weiyue Sun, Yingjing Xu, et al. MiniCPM-o 4.5: Towards real-time full-duplex omni-modal interaction. *arXiv preprint arXiv:2604.27393*, 2026.
- Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, and Neil Zeghidour. Moshi: a speech-text foundation model for real-time dialogue. *arXiv preprint arXiv:2410.00037*, 2024.
- Dujian Ding, Ankur Mallick, Chi Wang, Robert Sim, Subhabrata Mukherjee, Victor Rühle, Laks V. S. Lakshmanan, and Ahmed Hassan Awadallah. Hybrid LLM: Cost-efficient and quality-aware query routing. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2404.14618.
- Fahim Faisal, Sharlina Keshava, Md Mahfuz ibn Alam, and Antonios Anastasopoulos. SD-QA: Spoken dialectal question answering for the real world. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, 2021. arXiv:2109.12072.
- Qingkai Fang, Shoutao Guo, and Yang Feng. BayLing-Duplex: Native full-duplex speech dialogue with a single autoregressive LLM. *arXiv preprint arXiv:2606.14528*, 2026.
- Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630:625–630, 2024.
- Muye Huang, Lingling Zhang, Xingyu Yu, Lei Shi, Zhanyu Ma, Jun Xu, Jiuchong Gao, Jinghua Hao, Renqing He, and Jun Liu. DuplexOmni: Real-time listening, seeing, thinking, and speaking for full-duplex interaction. *arXiv preprint arXiv:2606.09186*, 2026.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.
- Jannik Kossen, Jiatong Han, Muhammed Razzak, Lisa Schut, Shreshth Malik, and Yarin Gal. Semantic entropy probes: Robust and cheap hallucination detection in LLMs. *arXiv preprint arXiv:2406.15927*, 2024.
- Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2302.09664.
- Tianpeng Li, Jun Liu, Tao Zhang, Yuanbo Fang, Da Pan, Mingrui Wang, et al. Baichuan-Audio: A unified framework for end-to-end speech interaction. *arXiv preprint arXiv:2502.17239*, 2025. Introduces OpenAudioBench.

Stephanie Lin, Jacob Hilton, and Owain Evans. Teaching models to express their uncertainty in words. *Transactions on Machine Learning Research*, 2022. arXiv:2205.14334.

William Lugoloobi, Thomas Foster, William Bankes, and Chris Russell. LLMs encode their failures: Predicting success from pre-generation activations. *arXiv preprint arXiv:2602.09924*, 2026.

Matéo Mahaut, Laura Aina, Paula Czarnowska, Momchil Hardalov, Thomas Müller, and Lluís Màrquez. Factual confidence of LLMs: on reliability and robustness of current estimators. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4554–4570, Bangkok, Thailand, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.250.

Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. In *Conference on Language Modeling (COLM)*, 2024. arXiv:2310.06824.

Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M. Waleed Kadous, and Ion Stoica. RouteLLM: Learning to route LLMs with preference data. *arXiv preprint arXiv:2406.18665*, 2024.

Hadas Orgad, Michael Toker, Zorik Gekhman, Roi Reichart, Idan Szpektor, Hadas Kotek, and Yonatan Belinkov. LLMs know more than they show: On the intrinsic representation of LLM hallucinations. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2410.02707.

Tanay Varshney, Annie Surla, Michelle Xu, Gomathy Venkata Krishnan, Maximilian Jeblick, David Austin, Neal Vaidya, and Davide Onofrio. LLM router: Rethinking routing with prefill activations. *arXiv preprint arXiv:2603.20895*, 2026.

Bandhav Veluri, Benjamin N. Peloquin, Bokai Yu, Hongyu Gong, and Shyamnath Gollakota. Beyond turn-based interfaces: Synchronous LLMs as full-duplex dialogue agents. In *Proceedings of EMNLP (Main)*, 2024. arXiv:2409.15594.

Peng Wang, Songshuo Lu, Yaohua Tang, Sijie Yan, Wei Xia, and Yuanjun Xiong. A full-duplex speech dialogue scheme based on large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024a. arXiv:2405.19487.

Xiong Wang, Yangze Li, Chaoyou Fu, Yunhang Shen, Lei Xie, Ke Li, Xing Sun, and Long Ma. Freeze-Omni: A smart and low latency speech-to-speech dialogue model with frozen LLM. *arXiv preprint arXiv:2411.00774*, 2024b.

Donghang Wu, Haoyang Zhang, Chen Chen, Tianyu Zhang, Fei Tian, Xuerui Yang, Gang Yu, Hexin Liu, Nana Hou, Yuchen Hu, and Eng Siong Chng. Chronological thinking in full-duplex spoken dialogue language models. In *Proceedings of the 27th Annual Meeting of the Special Interest Group on Discourse and Dialogue (SIGDIAL)*, 2026. arXiv:2510.05150.

Luoyuan Zhang, Bokai Xu, Junbo Cui, Weiyue Sun, Yingjing Xu, Hanyu Liu, and Yuan Yao. Liberating LLM capabilities in full-duplex speech models. *arXiv preprint arXiv:2606.07547*, 2026.

Xinrong Zhang, Yingfa Chen, Shengding Hu, Xu Han, Zihang Xu, Yuanwei Xu, Weilin Zhao, Maosong Sun, and Zhiyuan Liu. Beyond the turn-based game: Enabling real-time conversations with duplex models. In *Proceedings of EMNLP*, 2024. arXiv:2406.15718.

A Evaluation and calibration details

The Internal test set contains 240 queries: chat (60), factual QA (40), knowledge (60), math (60), and SimpleQA traps (20). Each query has unit weight, and every original query remains in the accuracy denominator. The four external English pools retain their original membership. Archived Mandarin diagnostics and AlpacaEval are excluded from the external macro average.

A.1 Experimental setup details

Models and decoding. The target is MiniCPM-o 4.5 (9B, based on Qwen3-8B). Controls include Qwen3-8B, MiniCPM-o 2.6, Qwen2.5-7B, Qwen2.5-Omni, Qwen2-Audio, and Freeze-Omni. Representation analyses use bf16, greedy decoding, and seed 42. Native decoding uses temperature $T = 0.7$ and top- $k = 20$. The expert is gpt-5.5 with low reasoning effort.

Probe fitting and thresholds. Representation probes use 360 calibration queries and single-position 4,096-dimensional features. The native gate uses 5,221 calibration rows and three L22 summaries: the last tail state, the tail mean, and the input mean. Its logistic head uses $C = 3 \times 10^{-4}$. Five-fold stratified cross-validation uses shuffle seed 42. For each language, conservative, balanced, and aggressive thresholds are the $(1 - r)$ quantiles of the 4,979 core rows’ out-of-fold scores, with $r \in \{0.15, 0.30, 0.50\}$. The 242 fresh training rows enter the fit but not the quantile base.

Judge configuration. Speech TriviaQA, Speech Web Questions, and Speech Llama Questions use the OpenAudioBench judge with gpt-4o-2024-08-06. Internal, SD-QA, and Mandarin reasoning answers use the reference-anchored gpt-5.4-mini adequacy judge. AlpacaEval uses VoiceBench’s gpt-4o-mini judge on its 1–5 scale. The adequacy judge receives the question, an optional reference, and a candidate answer without the system identity. Its rubric requires factual correctness, valid reasoning, and a complete answer. Prompt details are in Appendix E.

A.2 Metric provenance

Accuracy and escalation rates use judged native benchmark records. The expert receives at most the final 30 seconds of the prerecorded question. TTFA uses the complete ttfa-v3 run (run ID ttfa1) on the same query cohorts and named policies, with one attempt per query and policy. Inputs are replayed in real time, expert requests use the audio prefix consumed at the trigger, and both local and relay paths produce audio. The two sources are combined only at the policy-summary level. TTFA is available for 5,904 of 5,950 attempts; 64 completed early responses contribute zero wait. Candidate and stall audio and client playback are outside the endpoint.

A.3 Internal task strata

Table 2: Descriptive strata of the full 240-query Internal test set, with unit query weights. Local, aggressive (A), and always report answer-content accuracy (%); call rate is the aggressive arm’s realized rate. The final column counts question waveforms longer than the expert’s 30-second input window. All five categories and all original queries are retained.

Stratum	n	Local	A	Always	A call rate	> 30 s (n)
Chat	60	65.0	70.0	70.0	35.0	0
Factual QA	40	32.5	85.0	92.5	62.5	0
Knowledge	60	15.0	31.7	38.3	71.7	43
Math	60	61.7	61.7	78.3	25.0	3
SimpleQA traps	20	0.0	95.0	100.0	100.0	0

The largest gains occur on factual QA and SimpleQA traps. The fixed mix is not a production estimate; it preserves the full composition and every category.

B Representation and readout details

B.1 Layer and modality controls

Table 3 compares the duplex checkpoints with their raw backbones under the text-input LOPO-math protocol. Mid-layer transfer persists in both duplex models, whereas the final-layer decline is absent or smaller in the raw and streaming-omni controls.

Table 3: Text-input LOPO math AUC by depth for matched model families.

model	best mid-layer	final layer	depth profile
Qwen3-8B	0.964 (L33/36)	0.958	plateau
MiniCPM-o 4.5	0.931 (L22/36)	0.366	late decline
Qwen2.5-7B	0.893 (L21/28)	0.809	mild dip
Qwen2.5-Omni	0.794 (L16/28)	0.746	diffuse decline
MiniCPM-o 2.6	0.822 (L21/28)	0.540	late decline

The modality result is not limited to synthetic speech. On 200 query-disjoint human SD-QA recordings, the text-calibrated MiniCPM read reaches 0.760 AUC at L22 and 0.800 at L25. In the matched 600-question TTS experiment, L22 reaches 0.855. Because the evaluation sets differ, their gap is not an estimate of a human-speech penalty.

B.2 Mechanism checks

A speak-mode prompt template leaves final-layer LOPO math essentially unchanged (0.362 versus 0.366), the audio-text direction is nearly orthogonal to the failing probe, and source-pool identity remains decodable at all depths. Direct turn-control tests are also negative: the final probe’s maximum cosine with control-token unembeddings is 0.069 versus 0.068 for a random control, and a listen/speak prompt intervention gives cue-probe cosine 0.019 versus a 0.016 chance reference. We therefore use the depth result to choose a read location, without assigning the late-layer decline to a particular fine-tuning loss or training objective.

B.3 Native feature ablation

The matched ablation uses one native audio batch and identical training pools, labels, splits, and logistic-head recipe. Single-state and aggregated features have 4,096 and 12,288 dimensions. Each head fits on four pools and evaluates the fifth; the five-pool average includes Mandarin reasoning in addition to the four English pools. At a fixed 30% replay budget, routing gain combines recorded local and expert outcomes and compares with the analytical matched-random mix.

C Portability to a second duplex family

We replay NemotronLabs-VoiceChat-11B through its native frame protocol with the checkpoint frozen. Standardized logistic heads read L26/L30/L34 onset states; their normalized logits are averaged. The recipe is selected using calibration-only LOPO metrics. Its three-layer read reaches 0.818 out-of-fold AUC and 0.816 mean external AUC. The read covers the first 640 ms after commit, so this experiment tests method portability rather than a second end-to-end timing run.

On the 240-query Internal replay, 223 queries yield a valid onset read; the remaining 17 stay local. At nominal 15/30/50% scoreable-query budgets, the policy attains 31.7/44.6/60.4% accuracy, from a 19.6% local endpoint, while matched random reaches 29.6/39.9/53.5%. The layer set was reviewed using the Internal split, so we use the external LOPO result as the cleaner portability evidence.

D Gate operation and detailed results

The L22 hidden state is available after roughly 57–61% of prefill. On one H100, probe availability has P50/P95 of 20/25 ms for text and 45/104 ms for audio. These measurements cover the local read and exclude both the network transfer and the expert-model call.

Table 4: Native ablation by evaluation pool. Each column is one of the four matched LOPO readouts in Figure 4. The lower block gives accuracy gains over random escalation in percentage points. AUC averages weight pools equally; pooled routing gains weight individual questions.

Pool	L22		L35	
	Single	Aggregate	Single	Aggregate
<i>Failure-prediction AUC</i>				
Speech TriviaQA	0.645	0.744	0.704	0.661
Speech Web Questions	0.579	0.732	0.646	0.683
Speech Llama Questions	0.614	0.711	0.672	0.674
SD-QA	0.648	0.713	0.682	0.669
Reasoning QA (zh)	0.488	0.576	0.564	0.624
Equal-pool mean	0.595	0.695	0.654	0.662
<i>Accuracy gain over random at a 30% budget (pp)</i>				
Speech TriviaQA	+3.38	+4.58	+3.78	+3.38
Speech Web Questions	+1.69	+1.69	+4.16	+3.33
Speech Llama Questions	+1.51	+3.96	+2.73	+3.96
SD-QA	+2.85	+5.35	+5.85	+4.35
Reasoning QA (zh)	-3.91	-1.93	+1.04	+1.53
Pooled gain	+1.20	+2.78	+3.47	+3.33

Table 5: VoiceChat-11B cached native-frame replay, equally averaged over four external English speech-QA pools. Random uses the same call rate.

expert budget	15%	30%	50%
probe-guided accuracy	55.0	65.0	75.7
matched-random accuracy	50.7	57.7	67.0

Dialogue-act filtering. A second linear head uses the same L22 read to distinguish information requests from floor-management turns. Escalation requires both the information-seeking score $a(x)$ and failure score $s(x)$ to clear their thresholds:

$$d(x) = \mathbf{1}\{s(x) \geq \tau_\ell\} \mathbf{1}\{a(x) \geq \tau_a\}. \quad (2)$$

On the scripted calibration stimuli, the act head has out-of-fold AUC 1.000.

D.1 Realized expert-call rates

Table 6: Realized escalation rates (%) for MiniCPM native results in Table 1, plus the archived Mandarin and separate AlpacaEval pools. Internal retains all 240 queries, and every pool uses unit query weights. Rates can differ from nominal calibration and VoiceChat replay budgets.

Pool	n	Conservative	Balanced	Aggressive	Always
Internal	240	8.3	25.0	51.7	99.2
Speech TriviaQA	250	4.0	18.4	56.4	100.0
Speech WebQ	250	4.8	22.0	60.0	99.2
Speech Llama Q.	250	0.4	3.6	18.8	100.0
SD-QA	200	6.5	27.0	66.0	100.0
Reasoning QA (zh)	202	19.8	33.2	50.5	100.0
AlpacaEval	199	1.5	4.5	43.7	100.0

417 D.2 Internal accuracy and latency

Table 7: Internal results on all 240 original test queries (%), with unit query weights. Random mixes the measured local/always endpoints at the corresponding call rate. Accuracy has pointwise 95% percentile intervals from 100,000 paired query-bootstrap resamples (seed 42), conditional on the fixed test set and recorded arms. The intervals exclude judging and model-rerun uncertainty.

tier	calls	accuracy [95% CI]	random
local-only	0.0	40.8 [34.6,47.1]	—
conservative	8.3	46.7 [40.4,52.9]	43.3
balanced	25.0	54.2 [47.9,60.4]	48.2
aggressive	51.7	62.9 [56.7,68.8]	56.1
always	99.2	70.4 [64.6,76.3]	—

Table 8: Internal server TTFA (seconds), complete `ttfa-v3` run on all 240 test IDs: 240 attempts per arm, with 239/238/236/236/237 completed and 1/2/4/4/3 failed in row order. TTFA is the non-negative wait from scheduled input end to first answer PCM. All completed sessions count; the 17/12/11/13/7 early responses have zero wait. Failed attempts have no TTFA value. Timing ends at the first local PCM-bearing chunk return or relay PCM yield, excluding candidate/stall audio and client playback, which depends on the deployment environment.

tier	mean	P50	P95	P99
local-only	1.06	0.70	2.62	3.64
conservative	2.71	0.89	10.85	34.08
balanced	3.97	1.61	14.41	26.91
aggressive	6.10	2.87	18.14	54.78
always	9.43	6.67	25.86	60.83

418 D.3 Failure type and read time

419 A separate sweep fits the same readout family at successive native read points. Its five-pool macro
 420 includes Mandarin reasoning. Specific-but-wrong answers account for about 70% of failures and
 421 are ranked at AUC 0.813; perception errors reach 0.889, while execution slips reach 0.469. Reading
 422 two answer chunks later raises external macro AUC from 0.755 to 0.791, quantifying the trade-off
 423 between earlier intervention and information revealed during decoding.

Table 9: Failure-prediction AUC at successive native read points.

Read point	Calibration OOF	Internal	External macro
Before onset	.848	.832	.753
At onset	.850	.852	.755
After 1 chunk	.852	.854	.762
After 2 chunks	.855	.846	.791
After 3 chunks	.857	.858	.801

424 E Judging and reproducibility

425 The adequacy judge receives the question, optional reference, and candidate answer without the
 426 system identity. A response passes only when it is factually correct, uses valid reasoning, and
 427 completely answers the question; partial, hedged, refused, off-topic, and truncated answers fail.
 428 Representation-study sampling and splits use seed 42. Native benchmark arms use sampled decod-
 429 ing and one run per query. MiniCPM-o 4.5 runs with PyTorch 2.8.0 and Transformers 4.51.0 on
 430 single-H100 instances. The release includes calibration manifests, out-of-fold scores, thresholds,
 431 overlap exclusions, benchmark hashes, and the scripts needed to rebuild every reported table and
 432 figure from the recorded evaluation data.